

# Garmetix Sale, Purchase and Inventory Model / UI / Data-Layer Audit Report

Scope: Product, Stock, Invoice, PurchaseInvoice, BaseInvoice, InvoiceItem, PurchaseInvoiceItem, InvoicePayment, VendorPayment, CardPayment, Sale UI/API, Purchase UI/API, Inventory UI/API.

Prepared for the next implementation phase. This report identifies fields and relationships that are complete, missing, hidden, defaulted, or weakly linked.

**Overall conclusion:** the current system is a good starter workflow for basic POS sale, purchase inward, stock update, returns/exchange, printing, GST reports, and accounting hooks. It is not yet fully production-complete because several model fields are not exposed in UI, some relationships are inferred instead of stored, payment allocation is weak, GST split values are incomplete, and inventory has no movement ledger.

## 1. Executive Summary

| Area           | Current status                                                                                                    | Main gap                                                                                                                                                                |
|----------------|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sale           | Basic invoice creation, barcode item entry, print/PDF, return/exchange, stock deduction, loyalty earn.            | Missing payment split, card/UPI details, customer picker/advance/credit-note application, salesman, GST split, item snapshots.                                          |
| Purchase       | Purchase inward, new product on purchase, stock increment, print/PDF, full cancel/return, vendor payment voucher. | Purchase invoice lacks direct store fields, no partial item-wise return, weak purchase payment allocation, supplier bill date missing, freight GST/landed cost missing. |
| Inventory      | Product master, opening stock, stock list, barcode lookup, cache.                                                 | No stock movement ledger, no adjustment/transfer/count, limited product form, HSN only on stock, no concurrency guard.                                                  |
| Accounting/GST | Journal posting exists for main flows; GST module and bridge exist.                                               | Invoice/item CGST/SGST/IGST not fully populated, bill discount tax allocation not robust.                                                                               |

## 2. Product Model Findings

Product has Name, Barcode, MRP, TaxRate, Unit, TaxType, ProductType, Category, SubCategory, and stock navigation. UI only captures a small subset. Purchase inward can create products with defaults. HSN is not stored on Product in the original model; it is held at Stock level. For GST and stable invoice history, product/variant HSN should be available in the item snapshot and preferably in product master.

| Field / relation               | Status                         | Gap / action                                                     |
|--------------------------------|--------------------------------|------------------------------------------------------------------|
| Name, Barcode, MRP             | Handled                        | Used in Inventory, Billing, Purchase.                            |
| Description, Unit, ProductType | Weak                           | Mostly hidden/defaulted in UI.                                   |
| Category/SubCategory           | Partial                        | Purchase uses selectors; Inventory product form is limited.      |
| HSNCode                        | Missing on Product             | Stored on Stock only; invoice item snapshot should preserve HSN. |
| Brand/Style/Color/Vendor       | Model exists via ProductDetail | No complete UI flow in current sale/purchase/inventory.          |

## 3. Stock Model Findings

Stock stores cumulative PurchaseQty and SoldQty. This supports simple current stock calculation, but production inventory needs a movement ledger for audit, returns, transfers, adjustments, and reconciliation.

| Need                      | Current status                   | Risk                                                              |
|---------------------------|----------------------------------|-------------------------------------------------------------------|
| Purchase inward           | Increments PurchaseQty           | No movement row to audit each inward.                             |
| Sale outward              | Increments SoldQty and SoldValue | Concurrent sales can oversell without row lock/concurrency token. |
| Sales return              | Reduces SoldQty                  | No movement ledger, no reason/source line.                        |
| Purchase return/cancel    | Reduces PurchaseQty              | Full cancel only; item-wise purchase return missing.              |
| Adjustment/transfer/count | Missing                          | Cannot reconcile physical stock cleanly.                          |

## 4. Invoice and Item Findings

BaseInvoice has GST split fields but the implementation mainly calculates combined TaxAmount. InvoiceItem does not store historical ProductName, HSNCode, Unit, or GST split. This means old invoices can change appearance if product names/HSN are edited later.

| Model               | Handled                                                | Missing / weak                                                                               |
|---------------------|--------------------------------------------------------|----------------------------------------------------------------------------------------------|
| BaseInvoice         | Invoice number, status, totals, tax amount, round-off. | CGST/SGST/IGST not reliably populated; NetAmount meaning inconsistent.                       |
| Invoice             | Customer, store, items, payment, returns/exchange.     | Salesman not captured; B2B/B2C weak; advance/credit note/loyalty redemption not applied.     |
| InvoiceItem         | Product, barcode, qty, MRP, discount, tax %.           | No product-name snapshot, HSN snapshot, unit snapshot, category snapshot, or item GST split. |
| PurchaseInvoice     | Vendor, inward number/date, freight, status.           | No direct StoreId/StoreGroupId; supplier invoice date missing; item-wise return missing.     |
| PurchaseInvoiceItem | Reuses invoice item fields.                            | Inheritance hides Invoice navigation; not clean EF design.                                   |

## 5. Payment Model Findings

| Payment model    | Current use                                          | Gap                                                                                                          |
|------------------|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| InvoicePayment   | One sales payment row for paid sale/return/exchange. | Too simple: no BankAccountId, gateway/card/UPI/cheque detail, credit-note/advance source, settlement status. |
| VendorPayment    | Model exists but not aligned.                        | References sales Invoice, not PurchaseInvoice; not registered/used as purchase allocation.                   |
| CardPayment      | Model exists but not used.                           | Card number is int; should store masked last 4/reference, not full number.                                   |
| Purchase payment | Currently voucher/journal based.                     | Needs PurchasePayment / VendorPaymentAllocation table for correct paid/balance.                              |

## 6. UI / API Coverage Summary

| Module    | Complete / usable                                                                                     | Missing before production                                                                                                            |
|-----------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Sale      | Create invoice, barcode scan, stock deduction, print, sale return, exchange.                          | Customer picker, salesman, split payment, card/UPI detail, credit-note/advance apply, GST split, item snapshots.                     |
| Purchase  | Purchase inward, stock inward, new product, vendor GSTIN, print, full cancel, vendor payment voucher. | Store linked purchase invoice, supplier invoice date, partial purchase return, purchase payment allocation, freight tax/landed cost. |
| Inventory | Basic product and stock views, barcode lookup.                                                        | Full product master form, stock movement ledger, adjustment, transfer, physical count, min/max/reorder, concurrency.                 |

## 7. Priority Implementation Plan

- Phase 1 - Model alignment foundation: add purchase store fields, item snapshots, GST split fields, stock movement ledger, purchase payment allocation, payment detail extensions, product HSN field.
- Phase 2 - API/business logic: populate snapshots and GST split during sale/purchase/return/exchange, post stock movements, correct purchase paid/balance using allocation records, preserve original invoice values on cancellation.
- Phase 3 - UI completion: customer/vendor pickers, product full form, split payment form, card/UPI/cheque detail fields, advance/credit-note/loyalty redemption panels, partial purchase return UI.
- Phase 4 - hardening: sequence-safe invoice numbers, stock concurrency guard, bill discount tax allocation, stock adjustment/transfer/count, reports and tests.

## 8. Top Risks to Fix First

- 1 PurchaseInvoice is not directly linked to StoreGroup/Store.
- 2 VendorPayment/CardPayment models are not production-aligned or fully used.
- 3 Stock movements are not recorded as auditable rows.
- 4 Invoice items do not preserve product-name/HSN/unit/tax split snapshots.
- 5 GST split fields exist but are not reliably populated.
- 6 Invoice number generation and stock updates need concurrency protection.
- 7 Purchase return is full cancel only; item-wise purchase return is still required.

End of report.